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Ask of Koen

Proposal: Duration Tests (unified Markov)

Status: Proposed — unified source of truth (absorbs classroom usage + operator Console + infra/fleet execution)

Authors: Axle (infra/runner foundation); Steve (Lead Bot) (usage + operator Markov); unified 2026-09-30 per Koen + Steve agreement

Revision: 2026-09-30i — Koen feedback: drop initial scenario lists; per-state Return black circles; YAML for all layers; Implementation chapter; top-level Composite walks / How a walk becomes a test; UI Interactions hierarchy

Audience: Koen / IDEA leads

Companion capacity issue: [idea#159](#) — measure safe concurrent Kolibri video streams per instance

Backups (SUPERSEDED, content retained): [duration-tests-infra-backup.md](#) , [multi-engine-classroom.md](#) , [multi-engine-operator.md](#)

Purpose

Duration tests keep running to ensure **continuous school operation** over long time. Unit and integration tests prove isolated behaviours at a point in time; they do not catch emergent failures that only appear after sustained real-world use: stores diverging under repeated resync, instances in incorrect states after dock–undock–reboot sequences, CRDT conflicts accumulating over a day of school activity, or classroom/operator flows that only break after many hours.

A real school day involves ~6–8 hours of unpredictable events across 1–N Engines: teachers and learners using apps, operators managing disks/instances, and infra events (dock, move, reboot) with Automerge invariants checked throughout.

This proposal is the **single canonical** Markov model and execution design for those continuous tests. Prior separate drafts are backups only.

Conceptual model (Koen + Steve agreed)

One unified graph / one runner — three layers

Layer	What it covers	Entry from <code>start</code>
Usage	Classroom teacher / learner + apps (Kolibri, Nextcloud, Wikipedia/Kiwix)	Open Console as teacher / learner
Operator	Console manage/alter (dock/eject, install, start/stop, copy/move, backup/erase, operators, settings)	Open Console as operator → <code>op_entry</code>
Infra / fleet	Dock, move, reboot + Automerger invariants (YAML actions / runner)	One transition from <code>start</code> into the infra subgraph root (<code>infra_idle</code>)

Layers share one Markov walker and one Intent-style action naming convention. Usage and operator keep H2 state sections, actions unique to the departing state, and a UI Interactions chapter. The overall Markov graph is modeled as **YAML for all layers** (usage, operator, infra). Invariant specs are primarily for infra (optional light checks for usage/operator). Runner / stability / file-layout conclusions live in the **Implementation** chapter.

Return to start (every state)

Every state has a **Return to start** action so one walker can cover all three layer types in a single long run (leave usage → start → enter operator or infra, etc.).

Graphviz convention: do **not** draw large arrows from every state back to `start`, and do **not** use one shared black sink for all returns. Each state that returns gets its **own small dedicated filled black circle**, with a **short dashed shortcut arrow** from that state to its circle (edge label ≈ 0.05). Semantically the edge returns the walker to `start`; visually the per-state black circle cuts clutter versus long spokes into a common hub.

Placeholder $\approx$ probability for Return to start on each state: **0.05** (runner renormalises weights). Adjust when field frequencies exist.

Wiring rules (unchanged from prior drafts)

- **Actions are unique to the state they depart from** — you only take actions listed on the current state.
- **Arrival / entry UI** (login, open app, land on home) belongs on the **outgoing action of the state you leave**, not as content of the destination.
- Action name = edge label = UI Interaction name (Intent-style; no cryptic IDs; no `*` wildcards on the graph).

- A **real test** = a **random walk** for some duration or step count; each action expands into its UI Interaction (usage/operator) or infra action function (fleet).

Shared-store policy

Duration tests (and test/review deploys) must respect how Automerge documents and mDNS discovery interact with fleet isolation:

Context	Store document	Discovery (mDNS)	Notes
Single-Pi test / review	Unique doc (isolated store-url.txt / own store ID)	Off (IDEA_MDNS_DISABLE or equivalent)	Avoids contaminating peers; matches isolated Engine behaviour (AGENTS.md / idea#120, idea#155)
Multi-Engine test / review	Shared fleet doc	On	Exercises peer sync + NetworkTree mesh; Automerge convergence is a first-class invariant
Production (school)	Shared doc	On	Vision: Appdockers autofind; integrated catalog

Golden idea02 : protected from destructive churn. Never reboot, erase, or claim golden for duration-test actions. Infra actions that pick an Engine exclude role: golden (today idea02). See docs/PI_FLEET.md , idea fleet scripts, hw-roundtrip golden refusal.

Do **not** assume every fleet Pi always shares one live store during single-Pi work — that was a stale assumption in early duration-test drafts.

Vision cite

All Appdockers on the same LAN will automatically find one another and when a new Appdocker device is added, the catalog of available apps is automatically extended with the apps docked onto the new device.

Performance is optimized by adding Appdockers and redistributing the apps over the Appdockers.

Source: [solution-description.md](#)

Why multi-engine (Koen's three reasons)

1. Distribute apps / load across Engines

The physical act is simple: **add an Engine** and **plug a demanding app into it**.

- Engines **autofind and connect** on the school LAN (vision: Appdockers find one another).
- The **integrated app list** looks the same whether the school has 1 Engine or N Engines.
- Users **never need to know** which Engine serves which app, or which Console they opened — any Console on any Engine can see the full picture.

This is the same physical metaphor as the rest of IDEA: capacity is improved by adding hardware and redistributing disks, not by asking teachers to become sysadmins.

2. Scale one app across Engines

Some apps hit a **per-instance / per-Pi limit**. Example: concurrent Kolibri video streams.

When one Kolibri instance cannot safely serve the whole class:

- Start a **second Kolibri instance** on another Engine.
- Assign students** across instances (class A → instance 1, class B or half the cohort → instance 2).
- Same pattern later for Nextcloud video / heavy media.

Capacity baseline for Kolibri ([idea#159](#), idea01 Pi 5 / 4 GB). Multi-quality **planning** streams per instance (LAN measured SAFE ≥128 for all four; planning leaves Wi-Fi / browser / Pi-4 headroom):

Quality	Resolution	Planning streams / instance
low	360p (640×360)	64
medium	480p (854×480)	32
prior / typical lesson	720p (1280×720)	32
high	1080p (1920×1080)	12

Default classroom planning figure remains **32 @ ~720p**; use 64 / 12 when content is known low / high. Load-test tooling: [agent-app-dev#9](#).

3. Per-class Engine with own instances — Console manages everything from anywhere

Each class (or teaching block) can have **its own Engine** hosting **class-only** app instances — e.g. Class 5A Nextcloud, Class 5A Kolibri content — without isolating operators.

Any Console on any Engine can manage **all apps across all Engines** simply. Teachers and operators are not trapped on “their” Pi’s Console; the mesh presents one school.

Initial 3-engine setup (concrete sketch)

Design target first: a **named three-Engine school**, then generalise to N.

Name	Role (story)	Typical disks / instances
idea-A	Shared / school golden hub	Shared apps everyone uses lightly: offline Wikipedia (Kiwix), school-wide Nextcloud “School Files”, maybe a small shared Kolibri channel library. Stable reference Engine.
idea-B	Class Engine — Grade 5A	Class-only Kolibri (Grade 5A lessons/quizzes), class Nextcloud (Group “Grade 5A”: view-only shares, File Drop, collaborative docs).
idea-C	Class Engine — Form 3	Class-only Kolibri (Form 3), class Nextcloud (Group “Form 3”). When Grade 5A video load overflows idea-B, a second Kolibri instance for 5A can also land here (reason 2).

Classroom story: Monde Primary (or any Marco site) starts with one Engine in the room. When video buffering and “limit to one group at a time” become the norm (field troubleshooting already warns about many devices), the school adds **idea-B** and **idea-C** as classroom Engines. Teachers still open any Console; students still join `appnet` and pick apps from one list. Operators dock demanding class disks onto B/C and leave shared services on A.

Generalising to N

The three-Engine model is the **design target first**. N Engines later means:

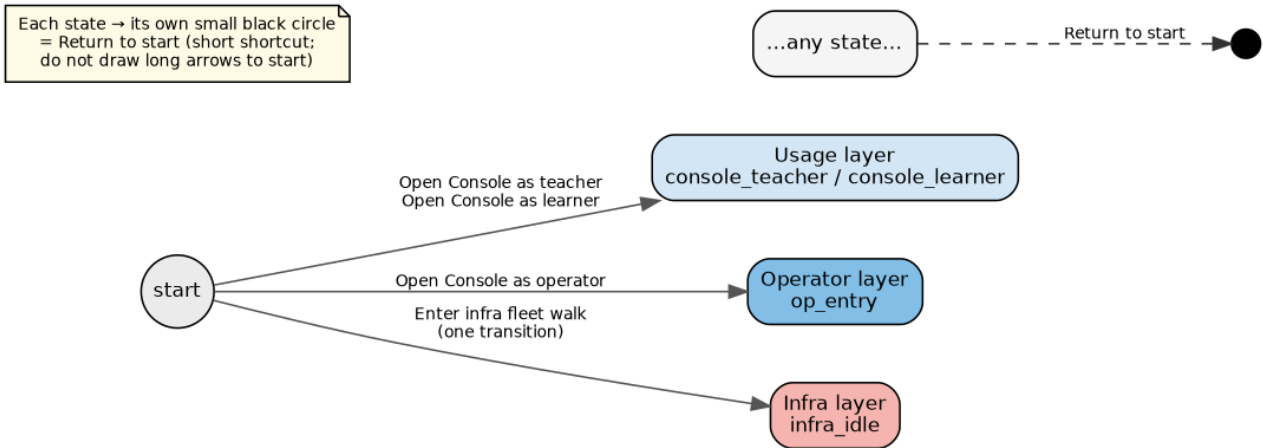
- More class Engines (idea-D...), and/or
- More replicas of a hot app (extra Kolibri / Nextcloud instances) assigned by cohort.

Discovery, Console overview, and “users don’t pick an Engine” stay the same rules; only inventory and assignment grow.

Unified graph overview

Diagrams are split into panels so Intent labels stay readable. Combined / panel sources:

Panel	File stem	Content
Hub	duration-tests-hub	start → usage / operator / infra roots; per-state Return black-circle convention legend
Usage A	duration-tests-usage-kolibri	Console + Kolibri (+ per-state Return shortcuts)
Usage B	duration-tests-usage-files	Console + Nextcloud + Wikipedia (+ per-state Return shortcuts)
Operator A	duration-tests-operator-hub	Auth, school hub, Account, Settings (+ per-state Return)
Operator B	duration-tests-operator-disk	Disk inventory and deep actions (+ per-state Return)
Infra	duration-tests-infra	Fleet dock / move / reboot subgraph (+ per-state Return)



Layer: Usage (classroom)

Classroom learner/teacher **usage** of Kolibri / Nextcloud / Wikipedia. Operator manage/alter is the next layer — not mixed into usage states.

Idea

- **States** = usage places (what a person is doing in an app or Console), not disk-dock hardware states alone.
- **Actions** = labeled transitions with probabilities (initial guesses until field frequencies exist).
- Test instances ship with **preloaded content**; legal actions derive from that content.
- Incorporate Marco's file-sharing and Kolibri classroom management, plus content-access for Kolibri, Nextcloud, and offline Wikipedia (Kiwix on idea-A).

States

State	Who	Colour	Meaning
console_teacher	Teacher	blue	School-wide Engine/app overview; opens apps as coach
console_learner	Student	mid-blue	Same unified app list; student opens an app to work
kolibri_home	Student or teacher	green	Kolibri signed in, Learn
kolibri_watching	Student	green	Playing an assigned lesson video
kolibri_exercise	Student	green	Doing an exercise in a lesson
kolibri_manage	Teacher	green (dark)	Classes, lessons, quizzes, Reports
nc_browse	Student or teacher	yellow	Nextcloud Files browse
nc_share	Teacher	yellow (dark)	View-only share to a class group
nc_drop	Student	yellow (dark)	Uploading to a File Drop
nc_collab	Student	yellow (dark)	Editing a shared collaborative doc
wiki_browse	Student or teacher	purple	Kiwix / offline Wikipedia on idea-A

Graphs

Diagram A — Console + Kolibri ([duration-tests-usage-kolibri.dot](#)):

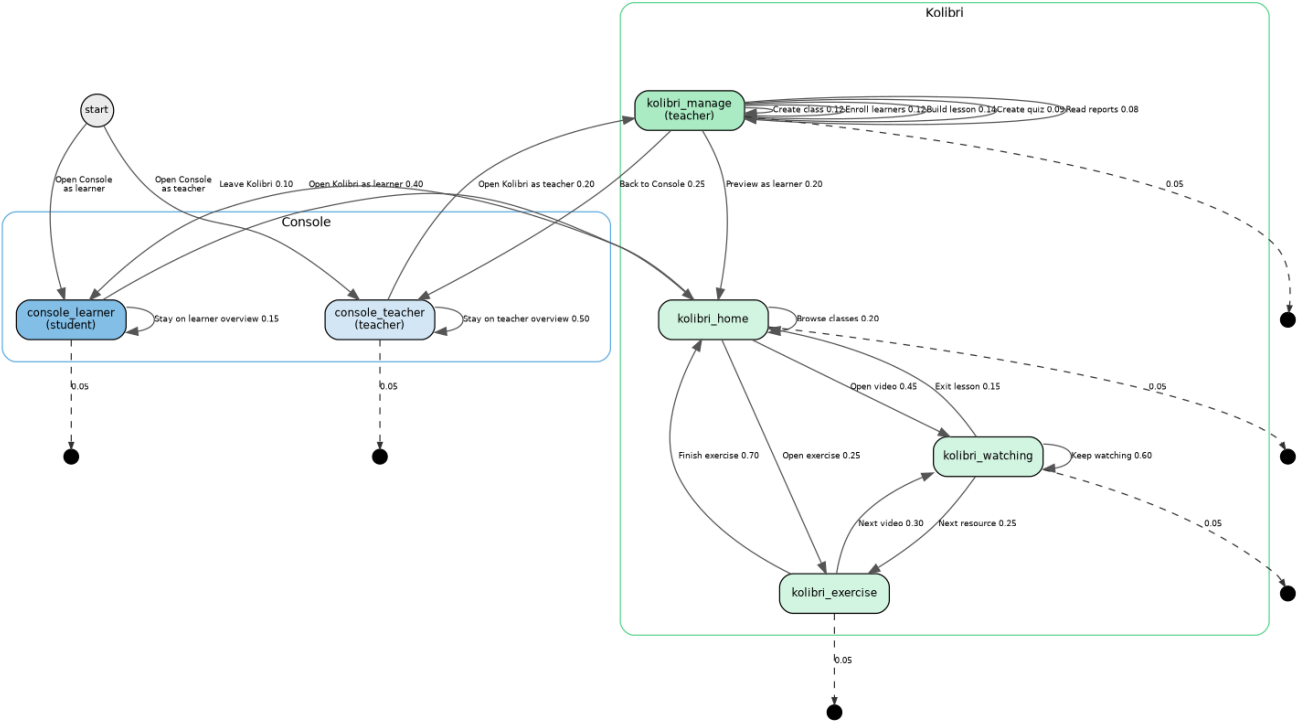
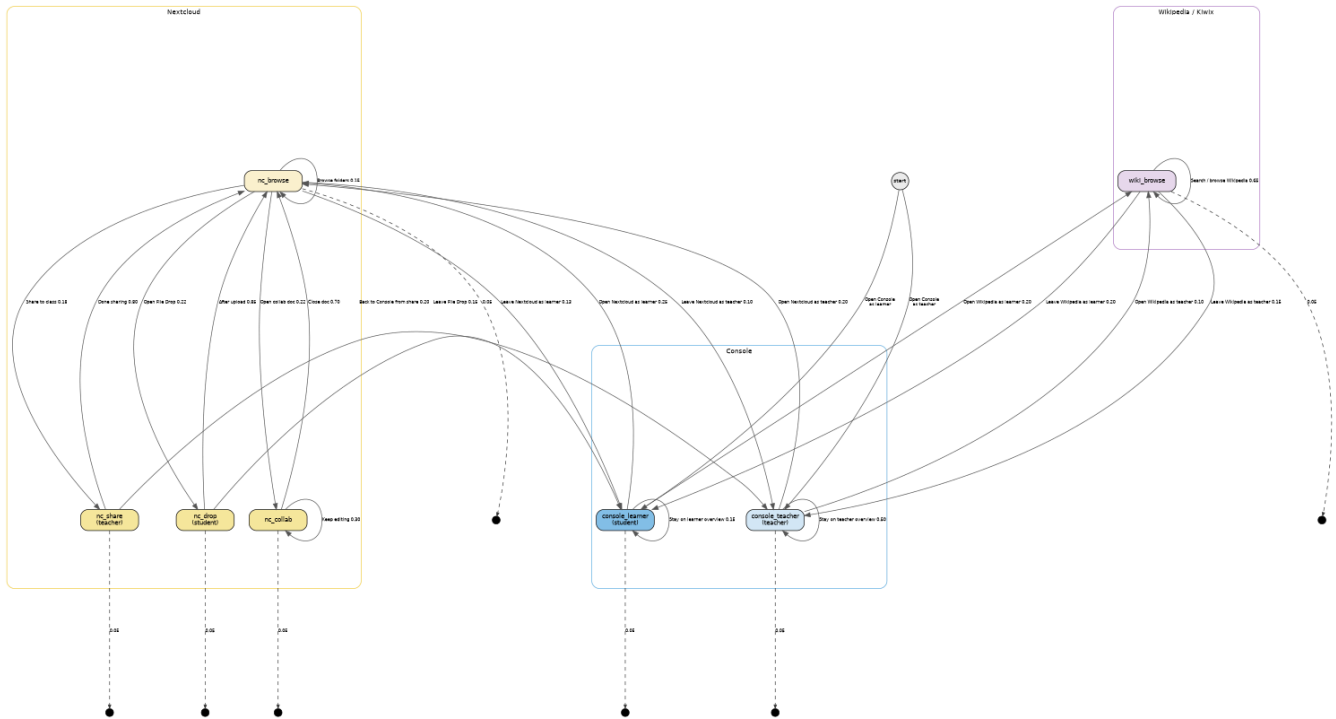


Diagram B — Console + Nextcloud + Wikipedia ([duration-tests-usage-files.dot](#)):



State: `console_teacher`

The teacher is on the **Engine / apps overview**: Engines present, apps docked, status. They are inspecting what is available on the network, not choosing a “home” Pi. **Operator-specific Console actions** (dock/eject disks, start/stop instances, install/move/backup/erase, operators/settings) are **not** on this usage graph — see [Layer: Operator](#).

While here: scan the integrated app list (Running, etc.) and Engine rows; optionally refresh / wait while students work.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open Kolibri as teacher	0.20	<code>kolibri_manage</code>	Open Kolibri as teacher
Open Nextcloud as teacher	0.20	<code>nc_browse</code>	Open Nextcloud as teacher
Open Wikipedia as teacher	0.10	<code>wiki_browse</code>	Open Wikipedia as teacher

Action	≈p	Destination	UI Interaction
Stay on teacher overview	0.50	console_teacher	Stay on teacher overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: console_learner

A student is on Console with the **same unified app list** as teachers (reason 1). They are not managing Engines — just picking Kolibri, Nextcloud, or Wikipedia.

While here: see integrated app list (class Kolibri/Nextcloud, Kiwix on idea-A, ...); choose an app or dwell.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open Kolibri as learner	0.40	kolibri_home	Open Kolibri as learner
Open Nextcloud as learner	0.25	nc_browse	Open Nextcloud as learner
Open Wikipedia as learner	0.20	wiki_browse	Open Wikipedia as learner
Stay on learner overview	0.15	console_learner	Stay on learner overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: kolibri_manage

The teacher is on Kolibri's **facility / coaching** side (Classes, Lessons, Quizzes, Reports) — not the learner Learn tab. Coaching Intents are **actions inside this state** (self-loops or leave).

While here: left sidebar visible; coach classes / lessons / quizzes / reports, or preview Learn / leave to Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
Create class	0.12	<code>kolibri_manage</code>	Create class
Enroll learners	0.12	<code>kolibri_manage</code>	Enroll learners
Build lesson	0.14	<code>kolibri_manage</code>	Build lesson
Create quiz	0.09	<code>kolibri_manage</code>	Create quiz
Read reports	0.08	<code>kolibri_manage</code>	Read reports
Preview as learner	0.20	<code>kolibri_home</code>	Preview as learner
Back to Console	0.25	<code>console_teacher</code>	Back to Console
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

Composite walks (Teacher prepares Grade 5A lesson, Quiz and reports) chain several of these — see [Composite walks](#).

State: `kolibri_home`

A student (or teacher previewing) is on the **learner** side after login — Learn tab, enrolled classes, assigned lessons.

While here: browse classes / lessons / quizzes; open a video or exercise; or leave Kolibri.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open video	0.45	<code>kolibri_watching</code>	Open video
Open exercise	0.25	<code>kolibri_exercise</code>	Open exercise
Browse classes	0.20	<code>kolibri_home</code>	Browse classes
Leave Kolibri	0.10	<code>console_learner</code>	Leave Kolibri

Action	≈p	Destination	UI Interaction
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `kolibri_watching`

Capacity-sensitive state (reason 2 / idea#159). Many concurrent walkers here is what forces a second Kolibri instance. Planning N depends on video quality: **64 @ 360p / 32 @ 480p–720p / 12 @ 1080p** (idea01 Pi 5 / 4 GB).

While here: video playing in-browser (no download); student may pause/seek; Kolibri records progress in the background.

Actions from this state:

Action	≈p	Destination	UI Interaction
Keep watching	0.60	<code>kolibri_watching</code>	Keep watching
Next resource	0.25	<code>kolibri_exercise</code>	Next resource
Exit lesson	0.15	<code>kolibri_home</code>	Exit lesson
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `kolibri_exercise`

While here: exercise questions with immediate feedback; student answers and submits; Kolibri records started / completed / score.

Actions from this state:

Action	≈p	Destination	UI Interaction
Finish exercise	0.70	<code>kolibri_home</code>	Finish exercise
Next video	0.30	<code>kolibri_watching</code>	Next video
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

(Quiz-taking can reuse exercise-like clicks under Learn → Quizzes, or stay under `kolibri_home` until a distinct load profile justifies a `kolibri_quiz` state. See [Composite walks](#).)

State: `nc_browse`

Hub for Marco's classroom file workflow. **Files** app is the central UI (manual p.12 in Marco's deck).

While here: browse folders (class materials, Drop Zone, school shares); start share / File Drop / collab; or leave to Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
Share to class	0.18	<code>nc_share</code>	Share to class
Open File Drop	0.22	<code>nc_drop</code>	Open File Drop
Open collab doc	0.22	<code>nc_collab</code>	Open collab doc
Browse folders	0.15	<code>nc_browse</code>	Browse folders
Leave Nextcloud as learner	0.13	<code>console_learner</code>	Leave Nextcloud as learner
Leave Nextcloud as teacher	0.10	<code>console_teacher</code>	Leave Nextcloud as teacher
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `nc_share`

While here: Share dialog open on a folder; set internal group share to View only; finish or leave.

Actions from this state:

Action	≈p	Destination	UI Interaction
Done sharing	0.80	<code>nc_browse</code>	Done sharing

Action	≈p	Destination	UI Interaction
Back to Console from share	0.20	console_teacher	Back to Console from share
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: nc_drop

While here: File Drop / file-request page open; upload file(s); then return to Files or leave to Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
After upload	0.85	nc_browse	After upload
Leave File Drop	0.15	console_learner	Leave File Drop
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: nc_collab

While here: collaborative editor open on a shared doc; peer avatars / live cursors; edit or close.

Actions from this state:

Action	≈p	Destination	UI Interaction
Close doc	0.70	nc_browse	Close doc
Keep editing	0.30	nc_collab	Keep editing
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: wiki_browse

Shared service on **idea-A** (school golden hub). Light load relative to Kolibri video — still a first-class classroom usage state.

While here: Kiwix library / Wikipedia ZIM open; search or browse articles; leave to matching Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
Search / browse Wikipedia	0.65	wiki_browse	Search / browse Wikipedia
Leave Wikipedia as learner	0.20	console_learner	Leave Wikipedia as learner
Leave Wikipedia as teacher	0.15	console_teacher	Leave Wikipedia as teacher
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

Layer: Operator (Console)

Authenticated operator **manage/alter** flows only. Classroom learner/teacher usage stays in the Usage layer.

Feature inventory (research — do not invent)

Inventoried from local clones `/workspace/agent-console-dev` and `/workspace/agent-engine-dev` (commands, components, docs). Only these land on the graph.

Console dual mode

Mode	Who	Login?	UI
User	students, teachers	No	AppBrowser — open running apps
Operator	IT / ops	Yes	NetworkTree + disk/instance management

Sources: `agent-console-dev/README.md` , `docs/ARCHITECTURE.md` , `proposals/console-user-management.md` .

Auth / operators (shipped)

Feature	UI	Notes
First-time setup	<code>FirstTimeSetup</code>	When <code>userDB</code> empty — create first admin
Log in	<code>LoginForm</code> / Account 	Username + password (bcrypt in <code>userDB</code>)
Log out	Account screen	Returns to user mode
Add / remove operators	<code>OperatorManagement</code>	Via Account → Manage Operators
Change password	Account / Settings → Account	Min 8 characters

School / multi-Engine overview (shipped)

Feature	UI	Notes
Network tree	<code>NetworkTree</code>	Engines + docked disks; role badges app / backup / files / empty / upgrade
Select Engine / disk	tree selection	Filters <code>InstanceList</code> / shows <code>DiskView</code> or <code>EmptyDiskPanel</code>
Engine discovery / connect	<code>ConnectionManagement</code> / Settings	Auto-discover hostnames; manual hostname; switch connected Engine
Reboot Engine	NetworkTree reboot control	Sends <code>reboot</code>
Operation progress / cancel	<code>OperationProgress</code>	<code>cancelOperation</code>
History / command log	<code>HistoryPanel</code> / <code>CommandHistory</code>	Observe traces

Disk roles & actions (shipped)

Disk role	Configure / act	Commands
Empty	Make Files Disk; Make Backup Disk; Install App; Erase	<code>createFilesDisk</code> , <code>createBackupDisk</code> , <code>installApp</code> , <code>summariseDisk</code> / <code>eraseDisk</code>

Disk role	Configure / act	Commands
App	Start/Stop/Open instances; Backup; Copy/Move; Add Files; Erase; Eject	<code>startInstance</code> , <code>stopInstance</code> , <code>backupApp</code> , <code>copyApp</code> , <code>moveApp</code> , <code>createFilesDisk</code> , <code>eraseDisk</code> , <code>ejectDisk</code>
Backup	Restore panel; (pure backup: no eject)	<code>restoreApp</code> ; <code>createBackupDisk</code> already done
Files	Share name / availability text; eject rules	<code>createFilesDisk</code> (add role); mount into opted-in apps is Engine-side
Upgrade	Badge only in tree	Upgrade ops appear in <code>operationDB</code> when an Upgrade Disk is processed — no separate Console “Upgrade” button inventoried
System	No eject / no erase	Protected

Instance actions (shipped — `InstanceRow`)

- **Start / Stop / Open** ↗
- **Backup** (picker when Backup Disks exist) → `backupApp`
- **Copy / Move** (drag-drop desktop; `MobileCopyMoveSheet` on mobile) → `copyApp` / `moveApp`

Dock / eject (shipped + physical)

Action	Where	Notes
Physical dock	USB plug into Engine	Disk appears in shared store / NetworkTree — not a Console click
Eject	NetworkTree / disk → <code>EjectConfirm</code>	<code>ejectDisk <diskId></code> — stops instances, unmounts; then operator unplugs
Undock (store)	Result of eject / physical removal	No separate “Undock” button beyond Eject

Explicitly not Console UI today (flagged)

Item	Status
Claim Engine / pool claim	Not a Console feature. Fleet claim scripts are test/Ops infra (<code>PI_FLEET.md</code>); classroom doc already lists them out of scope. Omitted from graph.
Dedicated uninstall / removeApp button	No Console command helper. Removing an app's instances is via erase disk (or stop + leave Stopped). Do not invent a remove-app state.
buildEngine provisioning	Engine CLI / Ops — not daily Console Markov.
Automatic “redistribute wizard”	Vision text only; Console redistribution = moveApp / copyApp + physical re-dock.

Confidence: High for commands in `src/store/commands.ts` and panels under `src/components/` listed above. Medium for Upgrade Disk operator *workflow* (badge + operation kinds exist; no dedicated upgrade click script). Low/N/A for claim — confirmed absent from Console.

Idea

- **States** = coarse operator places in the Console (not one state per click).
- **Actions** = manage transitions with placeholder probabilities. Intent-style explaining names only.
- Each action expands into a **UI Interaction** with the same Intent name.
- **Arrival UI rule:** open / confirm / land-on sequences belong on the **outgoing action of the state you leave**.
- **Actions are unique to the state they depart from.**

States

State	Meaning
<code>op_entry</code>	Open Console, connect/discover, Log in or first-time setup
<code>op_overview</code>	Operator NetworkTree — idea-A/B/C, disks, status (hub)
<code>op_disk</code>	Selected disk — EmptyDiskPanel or DiskView inventory
<code>op_eject</code>	Eject confirm dialog
<code>op_instance</code>	Start / Stop / Open / on-demand Backup on an instance

State	Meaning
op_install	Install App onto a disk
op_copy_move	Copy or Move app to another disk (possibly other Engine)
op_files	Make / Add Files Disk role
op_backup	Configure Backup Disk; Restore from Backup Disk
op_erase	Summarise + typed confirm erase
op_account	Operators list / add / remove / change password / logout
op_settings	Settings: Engine Connection, reboot, About

Graphs

Diagram A — Auth, school hub, Account, Settings ([duration-tests-operator-hub.dot](#)):

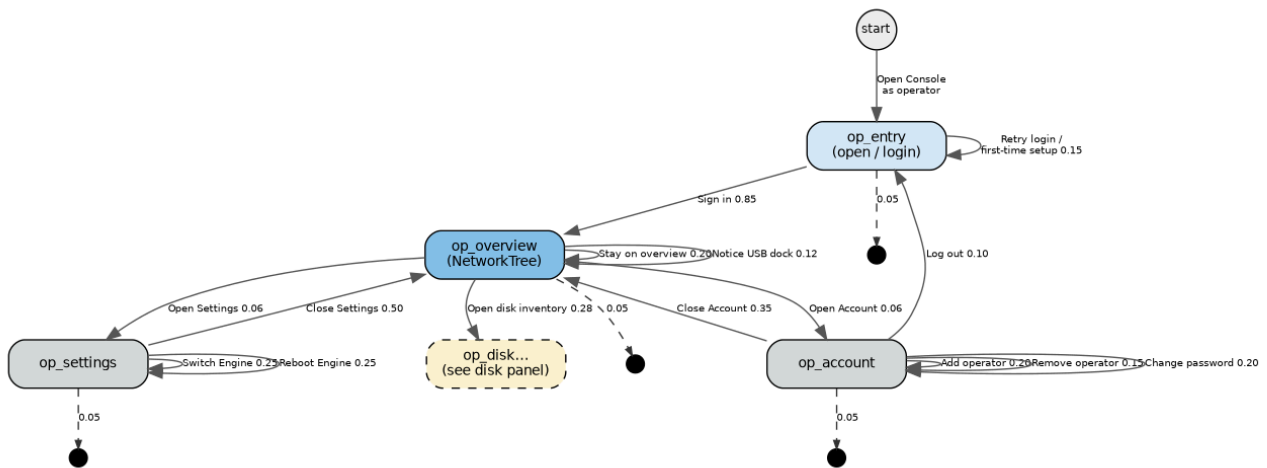
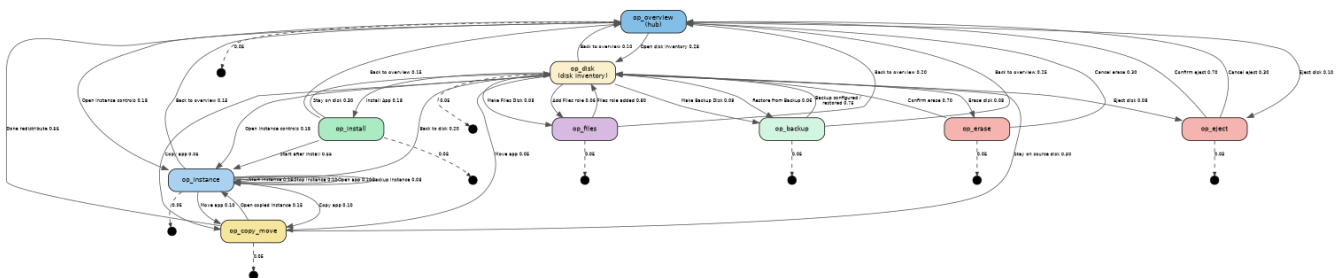


Diagram B — Disk inventory and deep actions ([duration-tests-operator-disk.dot](#)):



Multi-Engine story: operator may open Console on **any** of idea-A / idea-B / idea-C; after login the NetworkTree shows the **school mesh** when shared-store + discovery are on. Commands go to the Engine that owns the target disk (`dockedTo`).

State: `op_entry`

Operator is elevating into operator mode (or finishing first-time setup). Browser may already show user-mode AppBrowser.

While here: connect/discover if needed; enter credentials or create first admin; retry on failure.

Actions from this state:

Action	≈p	Destination	UI Interaction
Sign in	0.85	<code>op_overview</code>	Sign in
Retry login / first-time setup	0.15	<code>op_entry</code>	Retry login / first-time setup
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `op_overview`

Hub state. Operator sees Engines (idea-A / B / C), disks with role badges, instance status. Dwells, refreshes mentally as CRDT updates arrive, or reacts when a USB disk is docked.

While here: expand Engines in NetworkTree; note badges and status dots; open a surface (disk / instance / eject / Account / Settings) or dwell / notice USB dock.

Actions from this state (open surfaces only — not Install / Erase / Files / Backup):

Action	≈p	Destination	UI Interaction
Open disk inventory	0.28	<code>op_disk</code>	Open disk inventory
Open instance controls	0.18	<code>op_instance</code>	Open instance controls
Eject disk	0.10	<code>op_eject</code>	Eject disk

Action	≈p	Destination	UI Interaction
Open Account	0.06	op_account	Open Account
Open Settings	0.06	op_settings	Open Settings
Stay on overview	0.20	op_overview	Stay on overview
Notice USB dock	0.12	op_overview	Notice USB dock
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_disk

Empty disk → EmptyDiskPanel cards. App/Backup/Files disk → DiskView sections (Apps, Backups, Files). **Deep actions belong here.**

While here: choose Install / Files / Backup / instance / Copy-Move / Erase / Eject, or return to overview.

Actions from this state:

Action	≈p	Destination	UI Interaction
Install App	0.18	op_install	Install App
Make Files Disk	0.08	op_files	Make Files Disk
Add Files role	0.06	op_files	Add Files role
Make Backup Disk	0.08	op_backup	Make Backup Disk
Restore from Backup	0.06	op_backup	Restore from Backup
Open instance controls	0.18	op_instance	Open instance controls
Copy app	0.05	op_copy_move	Copy app
Move app	0.05	op_copy_move	Move app
Erase disk	0.08	op_erase	Erase disk
Eject disk	0.08	op_eject	Eject disk

Action	≈p	Destination	UI Interaction
Back to overview	0.10	op_overview	Back to overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_eject

While here: EjectConfirm dialog is open; confirm or cancel.

Actions from this state:

Action	≈p	Destination	UI Interaction
Confirm eject	0.70	op_overview	Confirm eject
Cancel eject	0.30	op_overview	Cancel eject
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_instance

On InstanceRow : Start, Stop, Open ↗, Backup (disk picker). Stay in-state for several toggles; leave to copy/move, disk, or overview.

While here: operate on the selected instance row.

Actions from this state:

Action	≈p	Destination	UI Interaction
Start instance	0.15	op_instance	Start instance
Stop instance	0.12	op_instance	Stop instance
Open app	0.10	op_instance	Open app
Backup instance	0.08	op_instance	Backup instance
Copy app	0.10	op_copy_move	Copy app

Action	≈p	Destination	UI Interaction
Move app	0.10	op_copy_move	Move app
Back to disk	0.20	op_disk	Back to disk
Back to overview	0.15	op_overview	Back to overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_install

While here: Install App flow in progress or just finished (catalog pick, OperationProgress); then start instance, stay on disk, or leave overview.

Actions from this state:

Action	≈p	Destination	UI Interaction
Start after install	0.55	op_instance	Start after install
Stay on disk	0.30	op_disk	Stay on disk
Back to overview	0.15	op_overview	Back to overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_copy_move

Vision “redistribute apps over Appdockers” = **Copy app / Move app** (+ eject/dock). Target disk must be docked to the Engine that receives the command.

While here: copy/move operation running or just finished; then overview, source disk, or open the new instance.

Actions from this state:

Action	≈p	Destination	UI Interaction
Done redistribute	0.55	op_overview	Done redistribute

Action	≈p	Destination	UI Interaction
Stay on source disk	0.30	op_disk	Stay on source disk
Open copied instance	0.15	op_instance	Open copied instance
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_files

While here: Make Files Disk / Add Files form (share name) submitted or in progress; then return to disk inventory or overview.

Actions from this state:

Action	≈p	Destination	UI Interaction
Files role added	0.80	op_disk	Files role added
Back to overview	0.20	op_overview	Back to overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: op_backup

While here: Backup Disk configure form or Restore panel active; then return to disk or overview. On-demand backupApp from an instance stays under op_instance (**Backup instance**).

Actions from this state:

Action	≈p	Destination	UI Interaction
Backup configured / restored	0.75	op_disk	Backup configured / restored
Back to overview	0.25	op_overview	Back to overview
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

State: `op_erase`

While here: `EraseDialog` open (summary shown); type exact label and confirm, or cancel.

Actions from this state:

Action	≈p	Destination	UI Interaction
Confirm erase	0.70	<code>op_disk</code>	Confirm erase
Cancel erase	0.30	<code>op_overview</code>	Cancel erase
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `op_account`

While here: Account / Manage Operators screen open.

Actions from this state:

Action	≈p	Destination	UI Interaction
Add operator	0.20	<code>op_account</code>	Add operator
Remove operator	0.15	<code>op_account</code>	Remove operator
Change password	0.20	<code>op_account</code>	Change password
Close Account	0.35	<code>op_overview</code>	Close Account
Log out	0.10	<code>op_entry</code>	Log out
Return to start	0.05	<code>start</code> (via dedicated black-circle shortcut)	Return to start

State: `op_settings`

While here: Settings panel open (Engine Connection / Account / About). Reboot may be modelled as a self-loop after `reboot` .

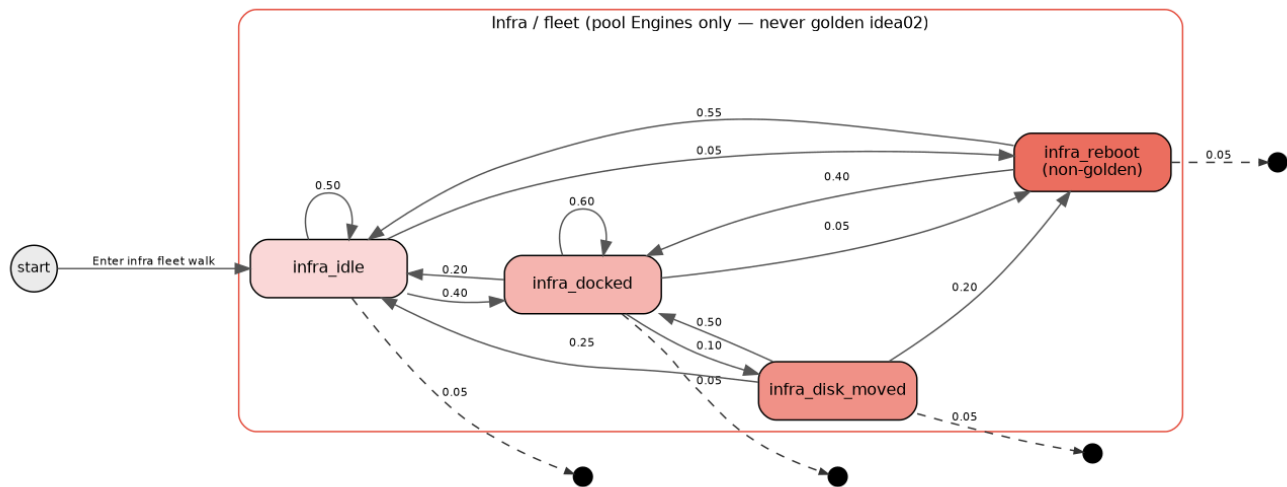
Actions from this state:

Action	≈p	Destination	UI Interaction
Close Settings	0.50	op_overview	Close Settings
Switch Engine	0.25	op_settings	Switch Engine
Reboot Engine	0.25	op_settings	Reboot Engine
Return to start	0.05	start (via dedicated black-circle shortcut)	Return to start

Layer: Infra / fleet (execution)

Fleet / harness actions that stress docking, disk moves, reboots, and **Automerge invariants**. Entered from `start` by **one** transition into the infra subgraph root `infra_idle` (Intent: **Enter infra fleet walk**). Scenario YAML and the walker cover **all** layers (see Scenario file below and [Implementation](#)); this layer's states/actions/invariants are the infra portion. Absorbs the former standalone duration-tests design ([duration-tests-infra-backup.md](#)).

Infra graph



State	Meaning	Action on entry
<code>infra_idle</code>	Engines up; no test disks docked (or undocked)	Undock fixture disks; verify instances Undocked

State	Meaning	Action on entry
<code>infra_docked</code>	Fixture App Disk docked on a chosen non-golden Engine	Dock fixture disk; wait settle
<code>infra_disk_moved</code>	Disk moved to a different Engine	Undock; re-dock on another pool Engine
<code>infra_reboot</code>	One non-golden Engine reboots / restarts	SSH reboot or <code>pm2 restart engine (--fast)</code>

Fleet isolation (current):

- Never target **golden** `idea02` for dock churn, erase, or reboot.
- Pool Engines only: `idea01` / `idea03` / `idea04` (or claimed spare/review roles).
- **Single-Pi** runs: unique store + discovery off — skip multi-Engine convergence asserts; still check local invariants.
- **Multi-Engine** runs: shared store + discovery on — require store convergence across participants.

Scenario file (YAML) — all layers

The **overall** Markov graph is modeled in YAML for **all** layers (usage, operator, infra) — not infra-only. One scenario file can mix layer entries from `start`. Intent names on transitions match UI Interactions (usage/operator) or infra action functions.

```
# scenarios/school-day.yaml
name: Typical school day
duration_minutes: 480 # 8 hours simulated; real time ~minutes via time compression
seed: 42 # optional RNG seed for reproducibility
exclude_engines: [idea02] # golden - never reboot / destructive infra

states:
  start:
    description: Neutral hub - sample layer entry
    transitions:
      - to: console_teacher
        weight: 25
        action: open_console_as_teacher
      - to: console_learner
        weight: 35
        action: open_console_as_learner
      - to: op_entry
        weight: 25
        action: open_console_as_operator
      - to: infra_idle
```

```
    weight: 15
    action: enter_infra_fleet_walk

# --- Usage (excerpt) ---
console_learner:
  layer: usage
  transitions:
    - to: kolibri_home
      weight: 40
      action: open_kolibri_as_learner
    - to: nc_browse
      weight: 25
      action: open_nextcloud_as_learner
    - to: wiki_browse
      weight: 20
      action: open_wikipedia_as_learner
    - to: console_learner
      weight: 15
      action: stay_on_learner_overview
    - to: start
      weight: 5
      action: return_to_start

kolibri_watching:
  layer: usage
  transitions:
    - to: kolibri_watching
      weight: 60
      action: keep_watching
    - to: kolibri_exercise
      weight: 25
      action: next_resource
    - to: kolibri_home
      weight: 15
      action: exit_lesson
    - to: start
      weight: 5
      action: return_to_start

# --- Operator (excerpt) ---
op_overview:
  layer: operator
  transitions:
    - to: op_disk
      weight: 28
      action: open_disk_inventory
    - to: op_instance
      weight: 18
      action: open_instance_controls
    - to: op_eject
      weight: 10
      action: eject_disk
```

```
- to: op_overview
  weight: 20
  action: stay_on_overview
- to: start
  weight: 5
  action: return_to_start

# --- Infra ---
infra_idle:
  layer: infra
  description: All participating engines up, fixture disks undocked
  transitions:
    - to: infra_docked
      weight: 40
    - to: infra_reboot
      weight: 5
    - to: infra_idle
      weight: 50
    - to: start
      weight: 5
      action: return_to_start

infra_docked:
  layer: infra
  description: Fixture disk docked on a pool engine
  transitions:
    - to: infra_idle
      weight: 20
    - to: infra_disk_moved
      weight: 10
    - to: infra_docked
      weight: 60
    - to: infra_reboot
      weight: 5
    - to: start
      weight: 5
      action: return_to_start

infra_disk_moved:
  layer: infra
  description: Disk physically moved to a different pool engine
  transitions:
    - to: infra_docked
      weight: 50
    - to: infra_idle
      weight: 25
    - to: infra_reboot
      weight: 20
    - to: start
      weight: 5
      action: return_to_start
```

```

infra_reboot:
  layer: infra
  description: One non-golden pool engine reboots
  transitions:
    - to: infra_idle
      weight: 55
    - to: infra_docked
      weight: 40
    - to: start
      weight: 5
      action: return_to_start

initial_state: start

```

Weights are relative — the runner normalises them. Full state tables in this document are the design source; YAML mirrors them for the walker. States map to **actions** (UI Interaction or infra function) and optional **invariants** after settling.

Invariant specs are **primarily for infra** (store convergence, phantom docks, zombie instances, golden untouched). Usage/operator may optionally declare light smoke asserts (e.g. catalog visible, NetworkTree present) — not required for every UI transition.

State actions (TypeScript)

State / Intent	Action
infra_idle	Undock fixture disks; verify instances Undocked
infra_docked	Dock fixture disk on chosen pool engine
infra_disk_moved	Undock from current; re-dock on a different pool engine
infra_reboot	SSH <code>sudo reboot</code> or <code>sudo pm2 restart engine (--fast)</code> on a non-golden engine
Return to start	Clear walker layer context; next step samples from <code>start</code>

Implemented as async functions in `test/duration/actions.ts` . YAML references action names; the runner resolves them.

Invariant verification

After each transition, wait for the system to **settle** (CRDT convergence when multi-Engine shared store) then verify invariants. **Primary focus: infra layer**. Usage/operator invariants are optional smoke checks only.

Global invariants (multi-Engine shared-store runs, after every infra transition):

- **Store convergence:** participating engines' `instanceDB` , `diskDB` , and `engineDB` match (CRDT equivalence)
- **No phantom docks:** no disk shows `dockedTo` pointing to an engine that no longer has the sentinel
- **No zombie instances:** no instance shows `Running` on a disk that is `Undocked`
- **Engine liveness:** engines that have not just been rebooted respond on WS within timeout
- **Golden untouched:** `idea02` (golden) not selected for destructive infra actions

Single-Pi / unique-doc runs: local variants of the above only (no cross-engine equality).

State-specific example:

```
infra_docked:
  invariants:
    - type: instance_status
      instance: kolibri-main
      expected: Running
    - type: disk_docked
      disk: test-app-disk
      engine: any_pool
```

CRDT convergence check

```
await waitForConvergence(fleetStores, timeoutMs)
```

Polls until serialised stores are equal or timeout (logged as convergence failure). Primary Automerge eventual-consistency check for multi-Engine runs.

Composite walks

Optional multi-action stories for discussion (not extra graph edges). Each step is a named UI Interaction (usage or operator) or infra Intent. Applies across **all** graphs / layers.

Usage

Walk	Chain
Teacher prepares Grade 5A lesson	Open Kolibri as teacher → Create class (<i>skip if preloaded</i>) → Enroll learners → Build lesson → optional Read reports → Preview as learner or Back to Console

Walk	Chain
Student completes assigned lesson	Open Console as learner → Open Kolibri as learner → Open video → Keep watching → Next resource → Finish exercise → Leave Kolibri
Quiz and reports	Create quiz → (<i>students: Learn → Quizzes → Start → answer</i>) → Read reports
Teacher share File Drop and collab	Open Nextcloud as teacher → Share to class → (<i>prep File Drop / create collab, often preload</i>) → students: Open File Drop / Open collab doc
Class hits stream ceiling	N × (Open Kolibri as learner → Open video → Keep watching); second instance start is operator

Operator

Walk	Story	Chain
Add capacity	New empty disk on idea-C → install Kolibri → start	Notice USB dock → Open disk inventory → Install App → Start after install
Redistribute load	Move heavy instance idea-B → idea-C	Open disk inventory → Move app → Done redistribute
Safe disk carry	Eject on idea-B → (unplug) → dock on idea-A	Eject disk → Confirm eject → Notice USB dock → Open disk inventory
Files for Nextcloud	Make Files Disk on idea-A → start Nextcloud	Make Files Disk → Files role added → Open instance controls → Start instance
Backup drill	Make Backup Disk → backupApp → restore elsewhere	Make Backup Disk → Backup configured / restored → Backup instance → Restore from Backup → Backup configured / restored

Infra (illustrative)

Walk	Chain
Dock–settle–undock	Enter infra fleet walk → (dock weights) → Return to start
Disk move across pool Engines	<code>infra_docked</code> → <code>infra_disk_moved</code> → <code>infra_docked</code> / <code>infra_idle</code>
Non-golden reboot drill	<code>infra_reboot</code> → <code>infra_idle</code> (assert golden untouched)

How a walk becomes a test

Applies to **all** layers — usage UI, operator UI, and infra actions — driven by the same Markov walker and scenario YAML.

1. Preload Kolibri with a class, enrolled learners, and a lesson that includes at least one video and one exercise (Marco Week 1–2 content shape).
2. Preload Nextcloud with class groups, a view-only materials folder, a File Drop folder, and one collaborative document (Marco classroom-use shape).
3. Preload / confirm Kiwix ZIM available on idea-A.
4. Sample a walk: e.g. `console_learner` → `kolibri_home` → ... → `kolibri_watching` (N concurrent walkers = N “students” hitting video — bounded by idea#159 quality table), or `console_teacher` → `kolibri_manage` for coaching scripts.
5. Multi-Engine walks assign walkers to instances on idea-B vs idea-C when testing reason 2; Console actions may start on any Engine’s Console (reason 3).
6. Operator walks preload three Engines in store (idea-A/B/C), an empty USB disk, an App Disk with a Stopped instance, and a Backup Disk when testing restore; each step expands to an Operator UI Interaction.
7. Infra walks use pool Engines only (never golden `idea02`); after each transition run infra invariants (and convergence when shared-store multi-Engine).
8. Any state may **Return to start** (dedicated black-circle shortcut) so one long run can leave usage, enter operator or infra, and continue.

Implementation of the walker, YAML loader, and action dispatch is in [Implementation](#).

Implementation

Follow-up implementation conclusions for the **unified** walker. Assessed: **yes — generalisation to all graphs is possible and intended**. One YAML schema, one runner, one action registry keyed by Intent name; usage/operator resolve to Playwright (or equivalent) UI Interactions, infra resolves to fleet harness functions. Layer tags on states (`usage` / `operator` / `infra`) select the dispatcher and which invariant suite runs. Return-to-start and shared-store / unique-doc mode switches are cross-cutting, not per-layer forks.

Test runner

```
pnpm test:duration [--scenario school-day] [--iterations 200] [--fast]
```

- `--scenario` : loads `scenarios/<name>.yaml`
- `--iterations` : transition count (default from YAML duration × avg dwell)
- `--fast` : `pm2 restart` instead of `reboot`; compresses dwell to seconds

Runner steps: parse YAML → connect to participating engines (`localStoreHandle` + WS, same idea as cross-engine tests) → walk Markov chain (usage UI / operator UI / infra actions) → structured log → on failure dump state + store snapshots + failing invariant.

Stability monitoring

During dwell, background probe ~every 30s: ping WS, `docker ps` for Running instances, unexpected status transitions. Logged; fail only after threshold (e.g. 3 consecutive failures).

File layout

```
test/
  duration/
    runner.ts      ← Markov walker + action dispatcher + invariant checker
    actions.ts     ← infra dock/undock/reboot (pool-only) + return_to_start
    ui/
      convergence.ts ← waitForConvergence + store equality checks
      invariants.ts  ← invariant type registry + evaluation
    scenarios/
      school-day.yaml ← default (may mix layer entries from start)
      stress.yaml     ← high-churn infra
      minimal.yaml    ← 2-engine / CI gate (non-golden)
```

Scenarios to ship

Scenario	Description	Focus	Duration
<code>school-day</code>	Typical day; usage + operator + infra from start hub	3 layers	8h simulated
<code>stress</code>	High reboot / disk-swap rate (pool only)	Infra	2h simulated
<code>minimal</code>	2 pool engines, small state space, CI gate	Infra (+ optional smoke usage)	10 min

Implementation phases

Phase 1: Runner + YAML loader + infra dock/undock/restart (pool-only) + convergence check + Return to start

Phase 2: Invariant registry + structured logs + shared-store / unique-doc mode switch

Phase 3: Wire usage + operator UI Interactions (Playwright) into the same walker

Phase 4: Stability monitoring + stress + CI `minimal`

Generalisation note: Phase 1–2 can ship infra-only YAML subsets; Phase 3 adds usage + operator states into the **same** schema without a second runner. Do not maintain parallel walkers per layer.

UI Interactions

Usage and operator **ordered UI click sequences**. Each name matches a graph action Intent. Markov actions link **forward** only; this chapter does not re-describe the graph.

Hierarchy: this section is the parent; **Usage UI Interactions** and **Operator UI Interactions** are the two subsections. Infra fleet steps are action functions (see Layer: Infra), not UI clicks here.

Usage and operator **ordered UI click sequences**. Each name matches a graph action Intent. Markov actions link **forward** only; this chapter does not re-describe the graph.

Return to start

Role: any walker · **From:** any state · **To:** `start` (via dedicated black-circle shortcut).

1. End the current UI / infra context (close dialogs if needed; do not leave the system dirty beyond what the departing state's invariants already allow).
2. Clear layer-specific walker context (signed-in app session may end; operator may remain elevated only if the next sampled entry expects it — default: return to neutral `start`).
3. Next transition is sampled from `start` (usage teacher/learner, operator entry, or **Enter infra fleet walk**).

Usage UI Interactions

Classroom teacher/learner UI click sequences. Intent-style names only (no cryptic IDs). Preload content so every click target exists.

Open Console as teacher

Role: teacher · **Engines:** any Console (idea-A / B / C equivalent).

1. Open Chromium (or any browser) on the school LAN → Console for any Engine (field habit today: hostname like `engine-1.local` ; multi-Engine story: idea-A / idea-B / idea-C Consoles must be equivalent).
2. Land on Engine / apps overview (Running apps, Engine rows).

Open Console as learner

Role: student · **Engines:** any Console.

1. Open browser on school LAN → any Engine's Console.
2. Land on unified app list (same catalog as teachers).

Stay on teacher overview

Role: teacher · Dwell on school-wide Console overview; refresh while students work; assert catalog visible without opening an app.

Stay on learner overview

Role: student · Dwell on unified app list without opening an app yet.

Open Kolibri as teacher

Role: teacher · Arrives in `kolibri_manage` .

1. From Console overview → click Kolibri (Running).
2. Kolibri login → type **teacher username** / **password** → Sign in.
3. Land on teacher home / facility UI (left sidebar visible) — ready for coaching actions.

Open Kolibri as learner

Role: student · Arrives in `kolibri_home` .

1. From Console overview → click Kolibri (Running).
2. Kolibri login → **learner username** / **password** → Sign in.
3. Top menu **Learn** → see enrolled classes / assigned Lessons (and Quizzes if active).

Open Nextcloud as teacher

Role: teacher · Arrives in `nc_browse` .

1. From Console overview → click Nextcloud (Running).
2. Nextcloud login → teacher username / password.

3. Open **Files** (default home) → ready to browse / share.

Open Nextcloud as learner

Role: student · Arrives in `nc_browse` .

1. From Console overview → click Nextcloud (Running).
2. Nextcloud login → student username / password.
3. Open **Files** → class materials / Drop Zone / shared docs.

Open Wikipedia as teacher

Role: teacher · Arrives in `wiki_browse` .

1. From Console overview → click **Kiwix** / Wikipedia app (Running on idea-A).
2. Kiwix library / Wikipedia ZIM opens in the browser.

Open Wikipedia as learner

Role: student · Arrives in `wiki_browse` .

1. From Console overview → click **Kiwix** / Wikipedia app (Running on idea-A).
2. Kiwix library / Wikipedia ZIM opens in the browser.

Create class

Role: teacher · *Requires already in* `kolibri_manage` .

1. Left sidebar **Classes** → **+ New class** → type name (e.g. `Grade 5A`) → **Save**.

Enroll learners

Role: teacher · *Requires class (preload or Create class).*

1. Open class → **Learners** tab → **Enroll learners** → tick students → **Confirm**.

Build lesson

Role: teacher

1. Class → **Lessons** → **+ New lesson** (or open existing) → name (e.g. `Week 1: Fractions Introduction`).
2. **Add resources** → Library/Channels → select **1 video** + **1 exercise** → **Save**.
3. Confirm **Recipients** = class → toggle lesson **Visible**.

Create quiz

Role: teacher

1. Class → **Quizzes** → **+ New quiz** → **Add questions** from exercise channel → set count (e.g. 5–10) → **Finish** → toggle **Active**.

Read reports

Role: teacher

1. Left sidebar **Reports** → **Classes** → class → **Lessons** or **Quizzes** → open item → scan learner table (scores / completion).

Preview as learner

Role: teacher · Leaves coaching for Learn (`kolibri_home`).

1. From coaching UI, open learner view / Learn (or sign in as a test learner).
2. See assigned lessons as a student would.

Back to Console

Role: teacher · Leaves Kolibri coaching for `console_teacher` .

1. Close / navigate away from Kolibri → Console overview visible again.

Open video

Role: student · From Learn / lesson list.

1. Open class → open assigned lesson.
2. Click video resource → player starts (enters `kolibri_watching`).

Open exercise

Role: student

1. Open class → open assigned lesson (or from lesson list).
2. Click exercise resource → exercise UI loads (enters `kolibri_exercise`).

Browse classes

Role: student · Stay in `kolibri_home` .

1. On Learn, open / close classes, scan assigned Lessons and Quizzes without starting a resource.

Leave Kolibri

Role: student · Returns to `console_learner` .

1. Leave Kolibri (close tab / navigate back) → Console unified app list.

Keep watching

Role: student · Self-loop in `kolibri_watching` .

1. Continue playing / pause / seek; Kolibri records progress.
2. *Concurrency note (S4):* N walkers dwelling here = N streams; if N > planning N for that quality on idea-B, second Kolibri on idea-C is an **operator** action.

Next resource

Role: student · Video → exercise in the same lesson.

1. Finish or leave video player → open the exercise resource in the lesson → `kolibri_exercise` .

Exit lesson

Role: student · Returns to `kolibri_home` .

1. Exit lesson / back to Learn or class list.

Finish exercise

Role: student · Returns to `kolibri_home` .

1. Answer remaining items → submit → return to Learn / class.

Next video

Role: student · Exercise → another video.

1. From exercise (or lesson list) open the next video resource → `kolibri_watching` .

Share to class

Role: teacher · Enters `nc_share` .

1. In Files, hover the folder (e.g. `Videos to watch`).

2. Click **Share** (person + icon).
3. Under **Internal shares**, type class group name (e.g. `Grade 5A`) → select group.
4. Set permission **View only**.

(Group must already exist — Accounts → Groups → + — often a preload step.)

Done sharing

Role: teacher · Returns to `nc_browse` .

1. Close Share dialog → back in Files browse.

Back to Console from share

Role: teacher · Leaves Nextcloud for `console_teacher` .

1. From Share / Files, leave Nextcloud → Console overview.

Open File Drop

Role: student (teacher prep optional) · Enters `nc_drop` .

1. *(Teacher prep / preload, often outside the walk:)* `Drop Zone` → Share → public link + → **File request** → copy link.
2. Student: open the File Drop / file-request link.
3. See empty “Click or drop to upload” UI (cannot see others' files).

After upload

Role: student · Returns to `nc_browse` .

1. Choose file(s) / drop → upload completes → return to Files browse.

Leave File Drop

Role: student · Returns to `console_learner` .

1. Leave File Drop page → Console.

Open collab doc

Role: student (teacher create often preload) · Enters `nc_collab` .

1. (Teacher create / preload:) Files → **+ New** → **New Document** → name → pick template → Share with group → **Allow editing**.
2. Student: open shared file from Files → editor loads; peer **avatars** top-right.

Close doc

Role: student · Returns to `nc_browse` .

1. Close editor → Files browse.

Keep editing

Role: student · Self-loop in `nc_collab` .

1. Continue typing / collaborating; cursors move live.

Browse folders

Role: student or teacher · Self-loop in `nc_browse` .

1. Navigate folders (class materials, Drop Zone, school shares) without starting share / drop / collab.

Leave Nextcloud as learner

Role: student · Returns to `console_learner` .

1. Leave Nextcloud → Console unified app list.

Leave Nextcloud as teacher

Role: teacher · Returns to `console_teacher` .

1. Leave Nextcloud → Console overview.

Search / browse Wikipedia

Role: learner or teacher · Self-loop in `wiki_browse` .

1. Search box: type a topic → open an article.
2. Follow internal links / browse categories; dwell reading.

Leave Wikipedia as learner

Role: student · Returns to `console_learner` .

1. Leave Kiwix → Console.

Leave Wikipedia as teacher

Role: teacher · Returns to `console_teacher` .

1. Leave Kiwix → Console.

Operator UI Interactions

Authenticated operator Console UI click sequences. Intent-style names only. Preload: three Engines visible in store (idea-A/B/C), at least one empty USB disk, one App Disk with a Stopped instance, one Backup Disk when testing restore.

Open Console as operator

Role: operator · **Engines:** any Console (idea-A / B / C).

1. Browser → Console URL on school LAN (or Chrome extension).
2. If needed: Settings → discover / Connect to an Engine (hostname).
3. Land in user-mode AppBrowser if not yet elevated — ready for **Sign in**.

Sign in

Role: operator · Enters `op_overview` .

1. Status bar  **Account** (or Log in) → Username / Password → **Log in**.
2. UI switches to operator layout: **NetworkTree** (left) + instances/disk detail.
3. Assert: NetworkTree visible with multiple Engines when mesh is up.

Retry login / first-time setup

Role: operator · Stays in / returns to `op_entry` .

1. If `userDB` empty → **First-time setup**: create admin username + password (min 8).
2. Or: failed login → correct credentials and retry Log in.
3. On success the next walk step is usually **Sign in** landing in overview (or setup completes into operator mode).

Stay on overview

1. Expand idea-A, idea-B, idea-C in NetworkTree.
2. Note disk badges (app / backup / files / empty) and instance status dots.

3. Dwell (CRDT updates) without changing selection.

Notice USB dock

1. From overview, note disk list.
2. (*Hardware / fleet harness*) Plug formatted or unformatted SSD into idea-C (or B).
3. Wait until new disk row appears under that Engine (still in `op_overview` ; next step often **Open disk inventory**).

Open disk inventory

1. Click an empty disk under idea-B → EmptyDiskPanel cards visible.
2. Or click an App Disk → DiskView Apps / Files / etc.
3. Arrives in `op_disk` .

Open instance controls

1. From overview or disk inventory, select an instance row (`InstanceRow` visible with Start / Stop / Open / Backup).
2. Arrives in `op_instance` .

Eject disk

Enters `op_eject` .

1. On a non-system, non-pure-backup disk with device: click **Eject**.
2. `EjectConfirm` dialog opens.

Confirm eject

1. Confirm in `EjectConfirm` .
2. Assert: disk undocked / removed from that Engine's docked list; instances stopped → back to `op_overview` .

Cancel eject

1. Cancel in `EjectConfirm` → back to `op_overview` (disk still docked).

Open Account

1.  → Account / Manage Operators screen → `op_account` .

Open Settings

1. ⚙️ → Settings panel (Engine Connection / Account / About) → `op_settings` .

Install App

Enters `op_install` .

1. Empty (or eligible) disk → **Install App** → pick app from catalog (e.g. Kolibri) → optional name → Install.
2. Wait `OperationProgress` .

Start after install

1. After install succeeds → **Start** → enter `op_instance` controls on the new instance.
2. Assert (reason 1): instance appears in every Console's catalog.

Stay on disk

1. After install (or from install state) remain on DiskView / EmptyDiskPanel without starting → `op_disk` .

Make Files Disk

Enters `op_files` .

1. Empty disk → **Make this a Files Disk** → share name (default `School Files`) → submit `createFilesDisk` .

Add Files role

Enters `op_files` .

1. App (or backup) Disk → **Add Files** → share name → submit.

Files role added

1. Assert: `files` badge; availability text updates when opted-in apps run → return to `op_disk` .

Make Backup Disk

Enters `op_backup` .

1. Empty disk → **Make this a Backup Disk** → mode on-demand / immediate / scheduled → select instance(s) → Configure.

Restore from Backup

Enters `op_backup` .

1. Select Backup Disk → Restore panel → pick instance archive → target App Disk → confirm Restore.

Backup configured / restored

1. Assert: `backup` badge (make) or restored instance present (restore) → return to `op_disk` .

Copy app

Enters `op_copy_move` .

1. From disk inventory or instance row on idea-B, **Copy** to App Disk on idea-C (drag or mobile sheet).
2. Wait `copyApp` operation → new InstanceID on target.

Move app

Enters `op_copy_move` .

1. **Move** demanding Kolibri from idea-A disk to idea-B disk (same InstanceID).
2. Wait `moveApp` ; assert backup links intact; source cleaned; catalog still unified.

Done redistribute

1. After copy/move completes → focus NetworkTree / school overview → `op_overview` .

Stay on source disk

1. After copy/move → remain on source DiskView → `op_disk` .

Open copied instance

1. Select the new (or moved) instance on the target disk → `op_instance` .

Erase disk

Enters `op_erase` .

1. From disk inventory: **Erase this disk...** → `EraseDialog` → wait `summariseDisk` summary.

Confirm erase

1. Type exact label → confirm `eraseDisk` .
2. Assert: disk `empty` ; prior instances gone → `op_disk` EmptyDiskPanel.

Cancel erase

1. Cancel `EraseDialog` → `op_overview` (or leave erase without wiping).

Start instance

1. Select Stopped instance on idea-B App Disk.
2. **Start** → wait until Running (or Error + History).

Stop instance

1. On Running instance → **Stop** → Stopped / Docked as applicable.

Open app

1. Running instance → **Open** ↗ → app URL loads (smoke only; deep app use is classroom graph).

Backup instance

1. Running (or eligible) instance → **Backup** → pick Backup Disk → `backupApp` .
2. Watch OperationProgress; cancel only if testing cancel path.

Back to disk

1. From instance controls, clear instance focus / select parent disk → `op_disk` .

Back to overview

1. Clear disk/instance selection or click school hub → NetworkTree overview → `op_overview` .

Add operator

1. Manage Operators → Add operator (username + password) → stay in `op_account` .

Remove operator

1. Manage Operators → Remove selected operator → stay in `op_account` .

Change password

1. Account → Change password (min 8) → stay in `op_account` .

Close Account

1. Close Account screen → `op_overview` .

Log out

1. Account → **Log out** → back to `op_entry` / user mode.

Close Settings

1. Close Settings panel → `op_overview` .

Switch Engine

1. ⚙️ → Engine Connection → Connect to another discovered Engine (or manual hostname) → stay in `op_settings` (tree refreshes for connected Engine).

Reboot Engine

1. NetworkTree **Reboot** on a selected Engine (modelled from settings/hub) → wait reconnect → stay in `op_settings` until **Close Settings**.

Sources consulted

Source	Use
<code>solution-description.md</code>	Vision: autofind Appdockers; redistribute for performance
<code>duration-tests-infra-backup.md</code>	Pre-unification infra YAML/runner (SUPERSEDED backup)
<code>multi-engine-classroom.md</code>	Pre-unification usage Markov (SUPERSEDED backup)
<code>multi-engine-operator.md</code>	Pre-unification operator Markov (SUPERSEDED backup)
<code>multi-engine-classroom-scenarios.md</code>	Older implementation-heavy draft (historical)
Marco presentations / quick-reference cards	Usage Intent / UI wording
<code>agent-console-dev</code> commands + components	Operator feature inventory

Source	Use
docs/PI_FLEET.md , idea fleet / golden rules	Fleet isolation; never churn idea02
idea#159	Kolibri concurrent stream capacity planning

Ask of Koen

1. Confirm **one** `duration-tests.md` as canonical; classroom/operator files remain backups only.
2. Confirm **Return to start** on every state via a **dedicated per-state black-circle** Graphviz shortcut (not one shared sink).
3. Confirm **shared-store policy** table (single-Pi unique+discovery off; multi-Engine/production shared+discovery on; golden protected).
4. Confirm infra entry is **one** transition from `start` into `infra_idle` .
5. Confirm scenario **YAML models all layers**; invariants primarily infra (optional usage/operator smoke).
6. Prioritise which layer / composite walks to deepen first for implementation (idea#85 lineage).